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Inherent Constant Current Constant Voltage (CC-CV) Charging of Electric Vehicles By Using LCL Resonant Converter

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***Abstract*—Due to its high efficiency and simple circuitry, the LLC Resonant converter is frequently utilized in onboard charger applications. To realize constant current (CC) and constant voltage (CV) charging, the typical LLC converter with variable frequency control suffers from a wide switching frequency range and also loose the zero-voltage switching (ZVS) at CC charging mode. In this research, a unique LCL resonant converter with built-in CC and CV output characteristics is offered as an alternative to the conventional solution to this issue. The CC and CV charges are implemented in the suggested concept employing the inherent properties of LCL resonant converter as a current source and as a voltage source. This paper also reports a design technique of the LCL resonant converter to achieve output voltage within the desired range in both CC and CV mode. The proposed converter may accomplish zero-voltage-switching (ZVS) for all primary-side inverter switches and zero-current-switching (ZCS) for load side rectifier diode during CC charge operation as well as during CV charge operation, this ensures a high efficiency. A 3.6-kW prototype with 250 V input and 126.3-168 V output is designed and simulated. The simulated results validate the proposed converter in CC-CV charging mode in ZVS region.**

Keywords— Constant current (CC) charging, constant voltage (CV) charging, LCL resonant converter, high efficiency, zero-voltage-switching (ZVS), zero-current-switching (ZCS).

I. Introduction

Due to their high levels of energy efficiency and environmental friendliness, PLUG-IN electric cars (PEVs) are becoming more and more popular [1]. The main element of PEVs is the on-board charger (OBC) [2]. The front-end PFC converter is cascaded by an isolated dc/dc converter in the conventional layout of the OBC because of the wide battery voltage range. To achieve high power factor (PF) and minimal input current harmonics, a front-end PFC converter is required. To perform the desired constant current (CC) and constant voltage (CV) charging for the battery with a wide voltage range (usually 126-168V), the post stage isolated dc/dc converter is used. A number of isolated dc-dc converters have been studied and can be divided into the following categories in order to achieve the isolated dc/dc converter's high conversion efficiency and required charging profiles [3].

The phase-shift PWM (PSPWM) converter is a topology that is frequently used in the first category of dc/dc converters because it operates at a fixed frequency and uses zero-voltage switching (ZVS) for main switches. The biggest drawbacks, however, are ZVS is lost at light load conditions. Moreover, circulating current and complex circuitry is another major problem in PSPWM converter. The turn-on loss could be minimized with the ZVS technique, but turn-off loss is still bound to the capability of the converter to work at a higher switching frequency. [4].

The second group makes use of DC/DC converters of the resonant kind. Amongst various dc-dc power conversion techniques, series resonant converter (SRC), parallel resonant converter (PRC) [5-8] and Series parallel resonant converter (SPRC/LCC) [9] are the three most popular topologies but all of these have some limitations. The LLC RC permits ZVS for the primary switches and zero-current switching (ZCS) for secondary diodes, resonant converter with variable frequency control becomes a well-liked choice [10]. The LLC resonant converter (LLC-RC) using variable frequency control (VFC) has a wide switching frequency range, which reduces efficiency and makes it more challenging to design magnetic components and EMI filters. A dual full bridge LLC converter [11] is provided as a solution to this problem in order to achieve fixed-frequency operation and high efficiency. However, this converter suffers following disadvantage including the complex circuitry and lower light load efficiency. Hybrid converters are an additional alternative. By combining a PSFB converter and an LLC converter, a hybrid converter is given [12]. As a result, the high efficiency, fixed frequency operation, and wide ZVS range are all concurrently achieved. The lagging leg's ZVS loss and the leading leg's large turn-off current are two disadvantage of this converter.

In conclusion, by using appropriate closed-loop control in conjunction with the battery voltage and current sense, the CC-CV charging characteristics of the abovementioned topologies are implemented. The CC-CV charging can be accomplished using special topologies with natural CC-CV features, which is another entirely distinct solution as compared to the control strategy-based CC-CV implementation technique. This technique reduces the sensing and control circuits while retaining wide ZVS region for primary switches [13], [14]. Thus high efficiency, high reliability, compact and low cost is the key feature of this converter. An LCL-T resonant converter [13] with clamp diodes is introduced. The natural CC output appears, in the LCL-T resonant converter. But clamp diodes are what basically do the CV charging. Because of the circulating current going back through clamp diodes to the input source, the CV charging stage's efficiency is consequently reduced.

In this paper, an unique LCL resonant topology is proposed in order to generate a topology with natural CC-CV features, and give the guaranty of ZVS, that is highly efficient and economically feasible. In this paper, the

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proposed converter's circuit configuration, theoretical analysis, optimized and design concerns are introduced. Analytical design and simulated a 3.36-kW prototype with a 250 V input and 126–168 V output. The simulated results show how effective and advanced the suggested converter is. The specifications of prototype are listed in table I.

TABLE I. SPECIFICATION OF THE PROTOTYPE

S. N.	Parameter	Value
1.	Input Voltage	250 V dc
2.	Rated Output Power	3360 W
3.	Rated Output Voltage	168 V dc
4.	Rated Output Current	20 A
5	Output current Range	20 – 7.6 A dc
6	Output Voltage Range	126 – 168 V dc
7	Switching Frequency Range	100 – 141.4 kHz

The paper is organized the following way. Section II describes the theory and analysis of the LCL resonant converter topology. The LCL-RC design have been discussed in sections III. Section IV includes results and discussion. Section V presents this paper's conclusion.

II. Theory & Analysis of LCL-Rc Topology As Cc-Cv Characteristics

The proposed converter is shown in Fig. 1. As shown in Fig. 1, the tank circuit of LCL-RC consists of passive resonant components L1, L2 in series, and C in parallel for getting the inherently CC-CV characteristics. The battery is modelled as an analogous resistance R_O, similar to the techniques in [15] – [17]. The switching network is similar to a square wave generator which generates a high-frequency square wave. The output of the switching network excites the LCL resonant tank circuit, (which in turn setup a resonant sinusoidal current) followed by a step down high-frequency transformer and full-wave center-tapped rectifier. The output of the rectifier contains ripples which is filtered by capacitor filters thus, output is smooth DC voltage.

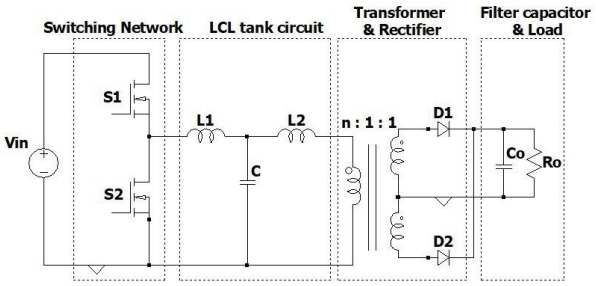

Fig. 1. Circuit diagram of the LLC resonant converter.

In the frequency domain method, the inverter's high-frequency component is omitted; hence, input excitation to the resonant circuit is considered a sinusoidal voltage. The only fundamental frequency component of the voltage and current is participating in the energy transfer and contribution of all other harmonic is neglected. The resonant tank circuit can be investigated using its first harmonic approximation (FHA) approach. Ac equivalent circuit of LCL RC, is shown in Fig. 2. In Fig. 2, the peak magnitude of the square wave input voltage is equal to the half of the input voltage (Vin/2) for half-bridge configuration.

Reflected load resistance (R_O') using the FHA technique, which is reflected on the primary side of the transformer, can be expressed as:

$$R_o' = \frac{V_o'}{I_o'} = \frac{\dfrac{2\sqrt{2}nV_o}{\pi}}{\dfrac{I_o\pi}{n2\sqrt{2}}} = \frac{8n^2R_o}{\pi^2} \tag{1}$$

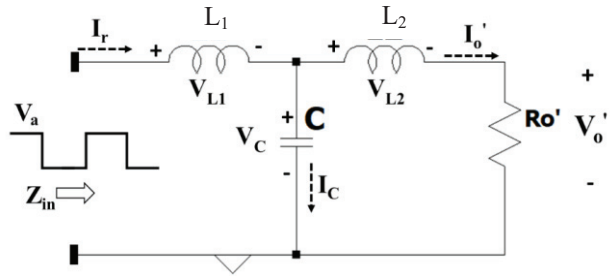

Fig. 2. A.C. equivalent circuit of LCL-RC.

Where $R_o = V_o/I_o$ and n = Np/Ns, Np, and Ns are the numbers of turns in primary and secondary winding, respectively.

The series resonant frequency (ω_o), quality factor (Q), normalized switching frequency (f_n or ω_n) and characteristic impedance (Z_o) are defined as:

$$\omega_O = \frac{1}{\sqrt{L_1 C}}, \quad Q = \frac{Z_O}{R_O'}, \quad \omega_n = \frac{\omega_S}{\omega_O} = \frac{f_S}{f_O} = f_n \quad \text{and} \quad Z_O = \sqrt{\frac{L_1}{C}}$$

where ω_s or f_S is switching frequency in rad/sec and Hz respectively and k is the ratio of inductors, i.e. ($k=L_2/L_1$).

$$V_0 = \sum_{1,3,5}^{\infty} \frac{2V_{dc}}{n\pi} \sin(n\omega t) \tag{2}$$

$$i_0 = \sum_{1,3,5}^{\infty} \frac{2V_{dc}}{n\pi Z_n} \sin(n\omega t - \phi_n) \tag{3}$$

v_o and i_o is the output of half bridge inverter and V_a is the rms voltage of fundamental component of v_o. Where $V_a = 1.414V_{in}/\pi$ and V_{in} is the input DC voltage.

The voltage gain (M) and current gain (H) are given as

$M=v_o'/v_a$ and $H= I_o/(v_a/z_o)$

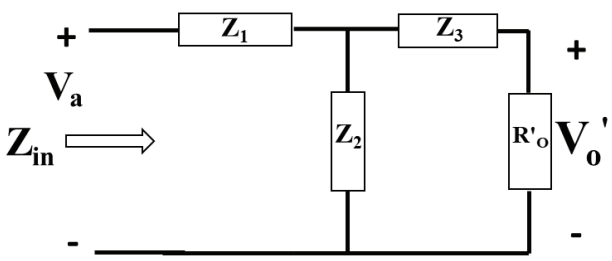

Fig. 3. A.C. equivalent circuit of LCL-RC

The output to input voltage gain [19] in the Laplace (s) domain can be expressed as:

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$$M(s)=\frac{V_o^{'}}{V_a}=\left|\frac{Z_2 \parallel (Z_a+R_o^{'})}{Z_2 \parallel (Z_a+R_o^{'})+Z_1}\times\frac{R_o^{'}}{Z_a+R_o^{'}}\right| \quad (4)$$

Where $Z_1 = sL_1, \quad Z_2 = \frac{1}{sC}, \quad Z_3 = sL_2$

Transferred from s domain to frequency domain by the equality of s = jω as:

$$M(s)=\frac{V_o^{'}}{V_a}=\left|\frac{1}{(1-\omega_n^2)+j\omega_n Q(1+k-\omega_n^2 k)}\right| \quad (5)$$

In Fig. 4, illustrates the plots (5) showing the voltage gain characteristics of the proposed converter with the variable load as a function of frequency. It can be observed from (5) that the voltage gain (M) is dependent on Q, f_n and the value of k. If you look closely at (5), you can see that the value of M is independent of the load (which is Q) if

$$j\omega_n Q(1+k-\omega_n^2 k)=0$$

$$\omega_n=\omega_v=\sqrt{\frac{1+k}{k}} \quad (6)$$

where ω_v is the normalized switching frequency where M is unaffected of Q. Moreover, the converter gain can be represented at $\omega_n = \omega_v$, as follows

$$M|_{\omega_n=\omega_v}=k \quad (7)$$

The current gain of LCL-T RC can be calculated using the following equation.

$$H=\left|\frac{Q}{(1-\omega_n^2)+j\omega_n Q(1+k-\omega_n^2 k)}\right| \quad (8)$$

we can observe from (8) that the current gain is not dependent on the load when operating the converter at $\omega_n = \omega_c = 1$ and at this frequency the current gain is

$$H|_{\omega_n=\omega_C}=1$$

Using the LTspice software, a parametric sweep analysis is carried out to determine voltage gain and current under various loading conditions in order to verify expressions (5) and (8). The operating frequency of the LCL-RC is also highlighted in this research. It can be observed in Fig. 4 that low Q curves belong to lighter load conditions while large Q curves belong to heavier load condition, also it can be

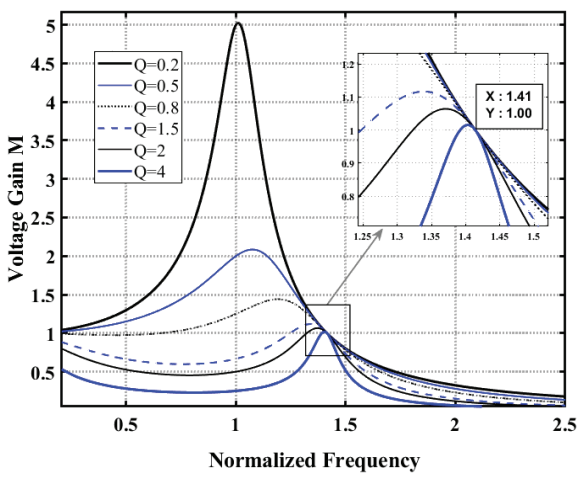

Fig. 4. Voltage gain characteristics and operating region at k = 1

observed that the peak gain point is moving with load. When the load condition is light, the peak gain point comes near the ω_c while at heavy load condition, this peak gain point move nearer ω_v. The voltage gain is constant and equal to k for all values of the quality factor at ω_v. Hence, at ω_v the voltage gain is not a function of output load resistance.

It can be observed in Fig. 5 the current gain (H) is constant and equal to 1 for all values of the quality factor at ω_c. Hence, at ω_c the current gain is not a function of output load resistance. Thus, after examine the Fig. 4 & Fig. 5 we can conclude that CC as well as CV output achieve, if switching frequency is equal to ω_c & ω_v respectively at all load conditions.

A steady-state analysis of this A.C. equivalent circuit is carried out to determine the magnitude of input impedance, phase angle and resonant frequencies.

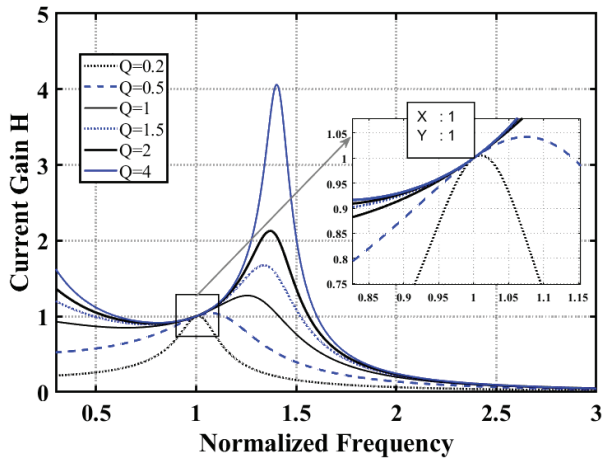

Fig. 5. Current gain characteristics and operating region at k = 1.

In Fig.3, the equivalent input impedance Z_{in} in s domain, normalized input impedance ($Z_n = Z_{in}/Z_o$) and phase angle (θ) between the input voltage and input current is derived as.

$$Z_{in}(s)=Z_1+Z_2 \parallel (Z_3+R_o^{'}) \quad (9)$$

$$Z_{in}(\omega)=j\omega L_1+\frac{\frac{1}{j\omega C_s}\times(j\omega L_2+R_o^{'})}{\frac{1}{j\omega C_s}+(j\omega L_2+R_o^{'})}$$

$$Z_n(\omega)=\frac{\left(\frac{1}{Q}+j\omega_n k\right)}{\left[(1-k\omega_n^2)+j\frac{\omega_n}{Q}\right]}+j\omega_n$$

$$\theta=\tan^{-1}\left(\frac{Imaginary\ Z_n(\omega)}{Real\ Z_n(\omega)}\right) \quad (10)$$

The plot of (10) that demonstrate the phase of the normalized input impedance of the proposed converter at k = 1, with the variable load as a function of frequency are shown in Fig. 6. For the frequency region below ω_c i.e. ($\omega_n < \omega_c$), where capacitor impedance dominates over inductor impedance for low Q and the converter operates in the ZCS region while for high Q, inductor impedance dominates over capacitor impedance, and the converter operates in the ZVS region. At ω_n above ω_v i.e. ($\omega_n > \omega_v$) θ will always be

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greater than equal to zero as a result, LCL-RC assures ZVS under all loading conditions. If the ω_n lies between ω_c and ω_v LCL-RC will work in ZVS or ZCS region determined by the load condition, opposite condition of above-mentioned frequency region ($\omega_n < \omega_c$).

The phase angle (θ) between the input voltage and input current at ($\omega_n = \omega_c$) is derived by simplified (6) as

$$\theta|_{\omega_n=\omega_c} = \tan^{-1}\big(Q(1-k)\big) \qquad (11)$$

$$\theta|_{\omega_n=\omega_c} = 0 \text{ , at k = 1}$$

$$\theta|_{\omega_n=\omega_c} \quad 0 \text{ , at k > 1}$$

$$\theta|_{\omega_n=\omega_c} \; 0, \quad \text{at k < 1} \qquad (12)$$

According to (12), LCL-RC can operate in inductive region (ZVS mode) if k < 1 (Fig.7). In spite of the fact that k = 1 results in a zero phase angle at ω_c and the lowest conduction loss, in reality, the necessary phase lag can be achieved by maintaining k slightly below unity. If k > 1 (Fig.8) LCL-RC operate in capacitive region at ω_c, and will lose ZVS mode, so proper care is needed in design of the LCL-RC for the choice of k.

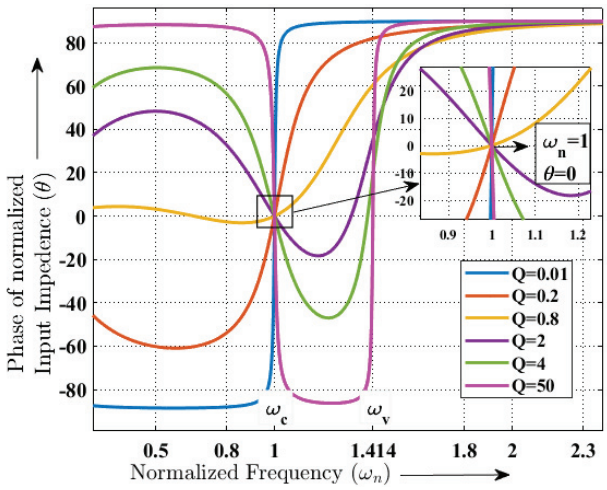

Fig. 6. Phase angle (θ), between input voltage (Va) and input current (Ir) as a function of ω_c at different values of Q and k = 1.

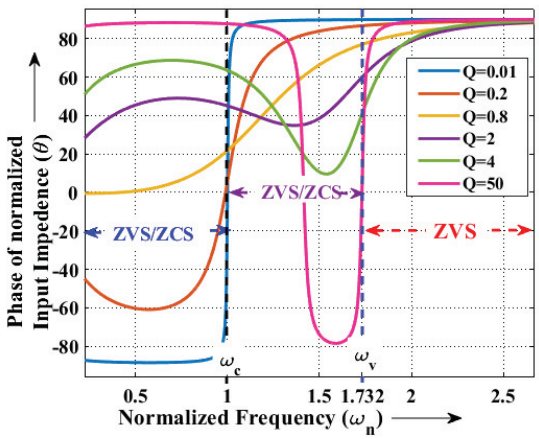

Fig. 7. Phase angle (θ), between input voltage (Va) and input current (Ir) as a function of ω_n at different values of Q and k = 0.5.

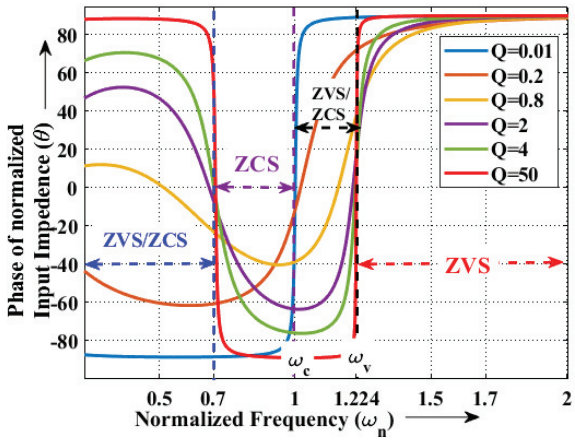

Fig. 8. Phase angle (θ), between input voltage (Va) and input current (Ir) as a function of ω_n at different values of Q and k = 2.

For application of CC-CV battery charging, MOSFET will be used as a primary switch, because MOSFET is the only device that can operate at a higher frequency. Thus, ZVS is preferred when MOSFET is used as a switch, under this switching mechanism, an ultra-fast anti-parallel diode is not required [18].

III. Design Of The LCL-Rc As Cc-Cv Characteristics

The voltage and current ratings of the reactive elements in the resonant networks were used to design the LCL-RCs. Q and k have a big impact on how well the reactive parts of the LCL resonant network design [19]. Determine the minimum operating frequency that will guarantee the converter must operate in CC mode and also in the inductive area for attaining ZVS at the maximum load condition. At all other load situations, the minimum operating frequency will also guarantee of CC and ZVS. According to a prior study, to achieve the ZVS condition, the value of k should be suitably less than one i.e., near to one. Therefore, considering the boundary point where the impedance must lie in the inductive region. So, for simulation purpose value of k has been chosen one.

As performance of battery charger in CC mode and CV mode depends on their respective gain (H & M). After the close examination of the equation (5) & (8) we found that voltage gain and current gain equal to unity at Q = 1, k = 1 and ωn = ωc. At Q = 1 voltage gain unity at both condition ωn = ωc as well as at ωn = ωv. Thus Q = 1 is only condition for getting desired output voltage at both CC and CV conditions.

Constant current (CC) charging and constant voltage (CV) charging stages are typically the two primary phases of the Li-Ion battery charging process. The output voltage of 168 V was planned for the battery charging system. As a result, 168 V was considered to be a reasonable charging voltage for the CV mode. A battery pack with a 6.1 kWh capacity was taken into account. The charge rate (C rate) of the battery pack can be estimated for 20 A of current in CC mode by

$$C = \frac{168\times 20}{6100} = 0.55 \qquad (13)$$

A. *Design of the resonant tank parameters*

After the calculation and simulation, we have decided the value of Q_{opt} and k is 1, and 1 respectively. First, we need to

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calculate the value of transformation ratio (n) then we require the find out the values of resonant tank components (L_1,C_1,L_2,and R_o').

The transformation ratio for the half-bridge switching circuit is given as:

$$n = \frac{|M_{nom}| \times V_{in}}{2(V_O + drop)} \tag{14}$$

$$C = \frac{1}{\omega_O{}^2 Q R_O'} \tag{15}$$

$$L_1 = \frac{Q R_O'}{\omega_O} \tag{16}$$

$$L_2 = kL_1 \tag{17}$$

Using (5), (8), (14), (15), (16), and (17), the following design values of various components are obtained. The designed value showed in Table II.

TABLE I. DESIGN PARAMETER OF THE LCL-RC.

S. N.	Parameter	Value
1	L_1	5.9 µH
2.	L_2	5.9 µH
3.	C	429.99 nF
4.	n	0.73
5	f_o	100 kHz
6	k	1
7	Q	1

IV. RESULTS AND DISCUSSION

Simulation of various circuits is done using MATLAB to understand the operating principle and behaviour of the LCL-RC to check the validity of the theoretical design. This section presents the simulation of LCL-RC in two operating modes, first mode is CC mode ($f_s = f_o$), and second mode is CV mode ($f_s = 1.414f_o$). Based on the design, an LCL resonant converter is developed with those parameters. The simulation result at different modes are shown in Fig. 9 - Fig. 13. Where the output voltage of the switching circuit (V_a), current in inductor L_1 (I_{L1}), output voltage (V_o), output power (P_o) and output current (I_o) is shown.

Only the following three load situations and three modes were taken into consideration because it is not possible to validate ZVS for every load condition. In MATLAB, the simulation results for the three below mentioned load conditions were plotted.

1) Start of the CC mode: In this mode, output current is 20 A, output voltage is 126.3 V, output power is 2.526 kW, and the load resistance is 6.315 Ω (Fig. 9).

2) The start of the CV mode (or the completion of the CC mode). In this instance, output current is 20 A, output voltage is 168 V and output power is 3.36 kW while the load resistance is 8.4 Ω (Fig. 10).

3) Termination of the CV mode: In this scenario, output current is 7.26 A, output voltage is 168 V, output power is 1.22 kW, and the load resistance is 23.13 Ω, (Fig. 11).

After the close examination of Fig. 9 & Fig. 10, converter maintain constant output current 20 A, while change in load resistance is 6.315 Ω to 8.4 Ω.

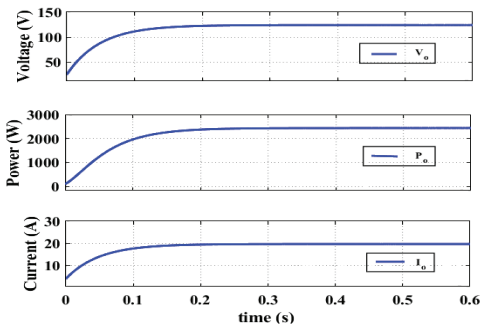

Fig. 9. Simulated waveforms of output voltage (Vo), output power (Po) and output current (Io) at start of CC mode, (R_o = 6.315 Ω).

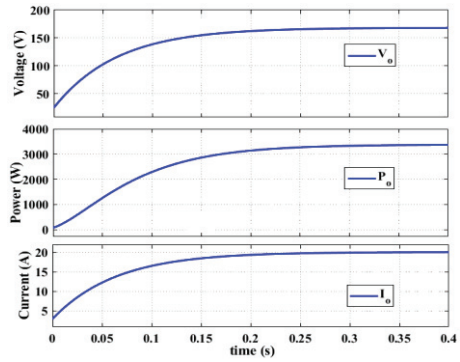

Fig. 10. Simulated waveforms of output voltage (Vo), output power (Po) and output current (Io) at end of CC mode, (R_o = 8.4 Ω).

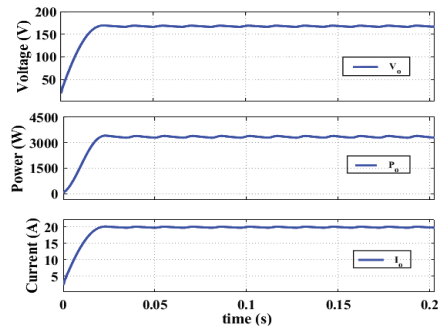

Fig. 11. Simulated waveforms of output voltage (Vo), output power (Po) and output current (Io) at start of CV mode, (R_o = 8.4 Ω).

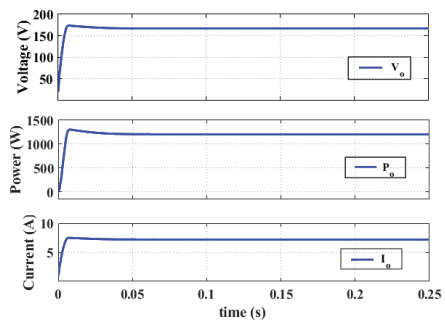

Fig. 12. Simulated waveforms of output voltage (Vo), output power (Po) and output current (Io) at termination of the CV mode (R_o = 23.13 Ω).

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After the close examination of Fig. 11 & Fig. 12, converter maintain constant output voltage 168 V, while change in load resistance is 8.4 Ω to 23.13 Ω.

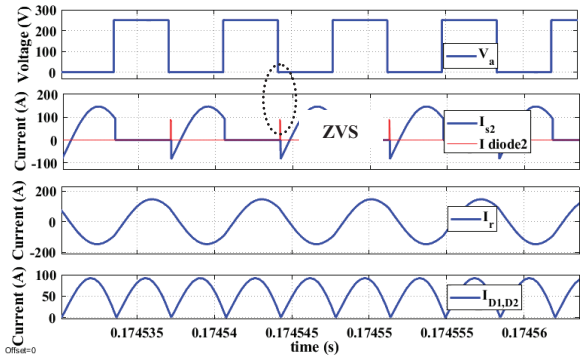

Fig. 13. Output voltage of inverter (V_a) (first), Drain current (I_{s2}) and antiparallel diode current (I_{diode2}) of MOSFET2 (second), resonant current of inductor L_1 (I_r) (third), output current of the full-wave rectifier ($I_{D1, D2}$) (fourth) at start of CV mode (R_o = 8.4 Ω).

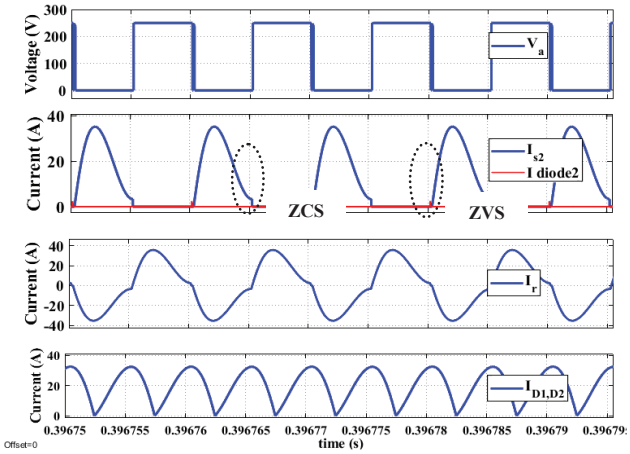

Fig. 14. Output voltage of inverter (V_a) (first), Drain current (I_{s2}) and antiparallel diode current (I_{diode2}) of MOSFET2 (second), resonant current of inductor L_1 (I_r) (third), output current of the full-wave rectifier ($I_{D1, D2}$) (fourth) at start of CC mode (R_o = 6.315 Ω).

V. Conslusion

In this paper, a brand-new LCL resonant converter for battery charging applications is suggested. It has natural CC-CV properties. In-depth descriptions are given of the circuit configuration, theoretical analysis, and design considerations. The proposed method and its validity for high power density are verified by the simulation using a 3.36 kW prototype with 250 V input and (168 V/ 20 A) rated output for charging of Plug-In Electric Vehicles. The circuit achieves ZVS from full load to light load condition, according to testing results from a 3.36kW prototype with 250V input and 126-168V output. Thus, it is simple to list the distinguishing characteristics of the suggested converter as follows.

1. The sensing and control circuit may not be necessary due to inherent CC-CV properties of proposed converter. As a result, the cost and reliability can be greatly improved.

2. Within the entire load range, the LCL resonant tank can guarantee ZVS operation for primary side switches and ZCS operation for secondary diodes. As a result, a high efficiency can be achieved.

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